

CASE FILE

PHISHING DETECTION & AWARENESS

>> EMAIL THREAT ANALYSIS & EMPLOYEE SECURITY AWARENESS

CASE REF# FUTURE-CS-02-2026
OPENED June 2026
LEAD ANALYST Angel Proshia F
SUBJECT Phishing Email Detection & Awareness
CLASSIFICATION CONFIDENTIAL
DOC VERSION 1.0

EXHIBITS LOGGED: 3

RISK PHISHING x3

FLAGS RAISED: 18

EVIDENCE
ANALYST REVIEW ONLY

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01 EXECUTIVE SUMMARY

This case file documents a phishing detection and awareness investigation conducted as part of the Future Interns Cyber Security Task 2 programme. Three email exhibits — modeled on real-world, publicly documented attack patterns — were examined using header forensics, sender domain verification, and link-destination tracing.

All three exhibits were classified as **PHISHING**. A total of 18 distinct indicators were logged across the three files, spanning sender spoofing, failed email authentication (SPF/DKIM/DMARC), urgency-based social engineering, and malicious link structures. Each finding below is explained in plain, non-technical language, followed by prevention guidance any organisation can hand directly to staff.

02 SCOPE & METHOD OF INVESTIGATION

OBJECTIVE	Examine phishing exhibits, log indicators, classify risk, issue staff briefing
EXHIBITS	3 email samples modeled on real-world phishing patterns
OPENED	June 2026
ANALYST	Angel Proshia F
TOOLS	Google Messageheader, MxToolbox Header Analyzer, browser link tracing
METHOD	Passive review only – no links clicked, no live infrastructure contacted

TOOL	PURPOSE IN THIS CASE
Google Admin Toolbox — Messageheader	Parses raw header text; verifies SPF / DKIM / DMARC authentication results
MxToolbox Email Header Analyzer	Cross-checks header authentication and traces the relay path
Browser Link Tracing	Safely previews link destinations without clicking
Google Docs / PDF	Compiles this case file into a client-ready document

03 RISK CLASSIFICATION KEY

Every exhibit reviewed in this case (and in any future review) is stamped with one of three classifications, defined below:

STAMP	MEANING	ACTION
SAFE	No indicators found; sender and links verified legitimate	Proceed normally
SUSPICIOUS	Unusual signs present but not conclusively malicious	Hold. Verify sender via a separate, known channel
PHISHING	Multiple confirmed indicators of a malicious or fraudulent message	Do not click, reply, or download. Escalate to IT/Security

04 HEADER FORENSICS (SPF / DKIM / DMARC)

Exhibit headers were run through two independent forensic tools — Google's Messageheader analyzer and MxToolbox's Email Header Analyzer — to check three authentication mechanisms. Both tools independently confirmed identical results, shown below.

CHECK	RESULT	FORENSIC MEANING
SPF	FAIL	Sending server not authorized to send for the claimed domain
DKIM	FAIL	No valid cryptographic signature — authenticity unverifiable
DMARC	FAIL	Domain policy explicitly flags this message as unauthenticated

VERIFIED VIA TWO INDEPENDENT TOOLS

Google Admin Toolbox Messageheader	CONFIRMED
MxToolbox Email Header Analyzer	CONFIRMED

Plain English: SPF, DKIM, and DMARC are technical checks that legitimate companies set up so receiving mail servers can verify a message really came from them. When all three fail together, it is one of the strongest possible technical signals that an email has been forged. Screenshots of both tool outputs are filed in screenshots/ and full header data is logged in evidence/email_header_analysis.md.

05 CASE FILES — EMAIL EXHIBITS

Each file below presents the exhibit's full message content, the flagged indicators identified during review, and an analyst's note explaining the attack in plain language.

FILE 01 Account Verification Scam

PHISHING

EXHIBIT 1 — RAW MESSAGE

From: Security Team
To: user@company.com
Subject: Urgent: Your Account Will Be Locked

Dear User,

We noticed suspicious activity on your account. To avoid account suspension, please verify your details immediately.

Verify Now: [http://secure-account-verify\[.\]com/login](http://secure-account-verify[.]com/login)

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

FLAGGED INDICATORS

- ✓ Generic greeting ("Dear User") instead of the recipient's real name
- ✓ Urgency and fear-based language ("suspicious activity", "24 hours", "permanent lock")
- ✓ Sender domain (secure-account-verify[.]com) does not match any real company
- ✓ Vague sender identity ("Security Team" — no company name given at all)
- ✓ SPF, DKIM, and DMARC authentication checks all FAILED (see header analysis)
- ✓ Call-to-action link designed to harvest login credentials

ANALYST NOTE

This is the most classic and recognisable type of phishing email — it works by creating panic. The message wants the reader to click without thinking, by threatening that their account will be locked. Real companies almost never lock an account within 24 hours over email alone, and they always use your real name, not "Dear User". The link goes to a fake login page designed to steal whatever username and password is typed in.

FILE 02 Invoice / Billing Scam (BEC-Style)

PHISHING

EXHIBIT 2 — RAW MESSAGE

From: Billing Department
To: finance@company.com
Subject: Re: Outstanding Invoice #INV-88341 — Payment Overdue

Hi,

This is a follow-up regarding invoice #INV-88341 which remains unpaid. Please find the attached invoice and process payment within 48 hours to avoid late fees and service interruption.

Download Invoice: [http://invoice-billing-secure\[.\]net/download?id=88341](http://invoice-billing-secure[.]net/download?id=88341)

If you have already paid, please disregard this message.

Best regards,
Billing Department
Accounts Receivable Team

FLAGGED INDICATORS

- ✓ Subject line uses “Re:” to falsely imply an ongoing conversation that never happened
- ✓ Targets a specific department (Finance) — a known Business Email Compromise (BEC) tactic
- ✓ Sender domain (invoice-billing-secure[.]net) is unrelated to any real vendor
- ✓ Urgency tied to money (“48 hours”, “late fees”, “service interruption”)
- ✓ Generic greeting (“Hi,”) with no specific name or real invoice context
- ✓ “Download Invoice” link likely leads to credential theft page or malware

ANALYST NOTE

This style targets people who handle company money, which is why it's called Business Email Compromise (BEC). It doesn't sound alarming or technical — it sounds like normal business correspondence, which is exactly why it's effective. Finance teams should always verify unexpected invoices directly with the vendor using a known phone number or contact, never by replying to the email or clicking the link inside it.

FILE 03

Fake IT Helpdesk Password Reset

PHISHING

EXHIBIT 3 – RAW MESSAGE

From: IT Helpdesk
To: employee@company.com
Subject: Action Required: Password Expires Today

Hello,

Our records show your network password expires today at 5:00 PM.
To avoid being locked out of your email and company systems, please
reset your password using the secure link below.

Reset Password: [http://company-support-portal\[.\]com/reset?user=employee](http://company-support-portal[.]com/reset?user=employee)

This link will expire in 2 hours for security reasons.

Thank you,
IT Helpdesk Team

FLAGGED INDICATORS	
✓	Impersonates an internal department (IT Helpdesk) using external domain
✓	Sender domain (company-support-portal[.]com) does not match the real company domain
✓	Extremely short urgency window (“2 hours”) to discourage careful checking
✓	Generic greeting (“Hello,”) despite claiming to have specific account records
✓	Link requests a password reset — classic credential harvesting technique
✓	No employee ID, ticket number, or other internal verification reference included

ANALYST NOTE

This is called spear phishing because it targets people inside one specific organisation by pretending to be their own IT department — a source employees are trained to trust and obey quickly. The short 2-hour deadline is designed to prevent the employee from pausing to verify the email with the real IT team. Legitimate IT departments almost always reset passwords through an internal company portal, never through an external link in an email.

06

DOMAIN & URL TRACE

All links across the three exhibits were traced using safe hover-preview inspection — no links were clicked and no live domains were visited. The table below compares the claimed sender identity against the actual linked domain.

FILE	CLAIMED SENDER	LINKED DOMAIN	VERDICT
01	Security Team	secure-account-verify[.]com	FAKE

02	Billing Department	invoice-billing-secure[.]net	FAKE
03	IT Helpdesk	company-support-portal[.]com	FAKE

- » None of the three links use the real, registered domain of the organisation they claim to represent.
- » All links use generic, business-sounding domain names invented specifically to appear official.
- » URL paths (/login, /download, /reset) mimic normal expected workflows to avoid suspicion.
- » No links were clicked during this investigation; findings are based on domain-name inspection only.

07 PREVENTION BRIEFING

The following briefing points are written for non-technical staff and can be used directly in security awareness training materials.

SLOW DOWN BEFORE CLICKING. Phishing relies on urgency. If an email pressures you to act within hours, treat that pressure itself as a warning sign.

CHECK THE SENDER'S ACTUAL DOMAIN. Don't just read the display name — hover over or tap the sender's email address to see the real domain behind it.

NEVER ENTER CREDENTIALS FROM AN EMAIL LINK. Instead, open a new browser tab and go to the company's website directly, or use a saved bookmark.

VERIFY UNUSUAL REQUESTS THROUGH A SECOND CHANNEL. If an email asks for payment, login, or sensitive data, confirm it by phone or in person before acting.

REPORT, DON'T JUST DELETE. Forward suspicious emails to your IT/Security team so they can warn other employees and block the sender.

08 FIELD GUIDE — DO'S AND DON'TS

DO	DON'T
Verify sender domains before clicking any link	Click links in unexpected or urgent-sounding emails
Report suspicious emails to IT/Security immediately	Reply to or forward suspicious emails to coworkers
Type known company URLs directly into your browser	Enter your password on a page reached via an email link
Confirm payment or data requests via phone	Process payments based on email instructions alone
Use multi-factor authentication wherever available	Reuse the same password across multiple accounts

09

CHAIN OF CUSTODY / ETHICS

This investigation was conducted **solely for educational purposes** as part of the Future Interns Cyber Security programme. The phishing exhibits examined in this case file are either drawn from the task brief or modeled on common, publicly documented phishing patterns. All domains referenced are defanged (written with [.] instead of .) and are illustrative placeholders — no real organisation, live website, or real victim was involved.

- ✓ No links were clicked; no attachments were opened; no live malicious infrastructure was contacted
- ✓ All sample emails were either provided in the task brief or constructed to illustrate publicly known phishing patterns
- ✓ Email header data shown is illustrative, built to demonstrate the SPF/DKIM/DMARC analysis process
- ✓ No real personal, financial, or organisational data is included anywhere in this file
- ✓ Full compliance with Future Interns Task 2 ethical guidelines

CASE REF# FUTURE -CS-02-2026	ANALYST Angel Proshia F	DATE June 2026	VERSION 1.0 – FINAL	CLASS. CONFIDENTIAL
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